

10 (a) Simplify $(2p)^9$ where $p > 0$

(1)

$$y^a \times y^b = y^c$$

(b) Find the value of n

$$\begin{aligned} n &= 9 + -3 \\ &= 9 - 3 = \underline{\underline{6}} \end{aligned}$$

$$n = 6$$

(1)

(c) Simplify fully $(5a^4c^2)^3$

$$\begin{aligned} &5^3 a^{4 \times 3} c^{2 \times 3} \\ &= \underline{\underline{125 a^{12} c^6}} \end{aligned}$$

$$125 a^{12} c^6$$

(2)

(Total for Question 10 is 4 marks)